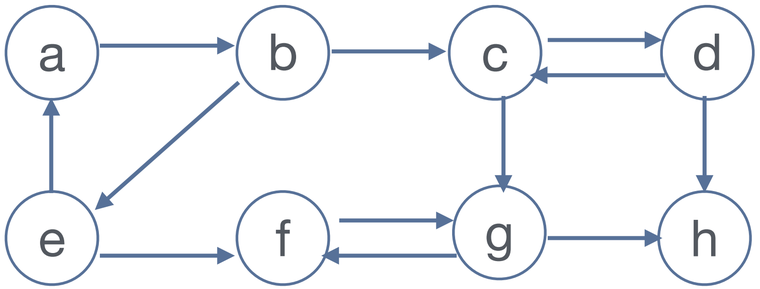
In-Class Assignment L4-1 (Group of 4 or less): Strongly-Connected Component

(20 minutes) Submit to LEB2.

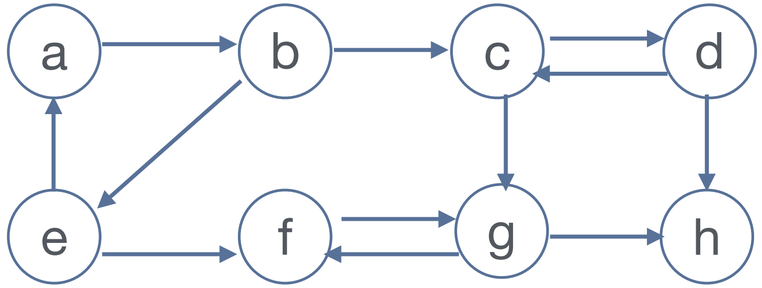
For the following digraph, determine all SCCs by using Kosaraju–Sharir algorithm.

- Show the node order used to traverse the transposed graph G^T and individual SCCs obtained from traversing G^T.
- Show how the SCCs are connected with the acyclic component graph.



Solution

G



$h_{5,1}$

$d_{4,2}$ $f_{7,3}$

$c_{3,5}$ $g_{6,4}$ $e_{8,6}$

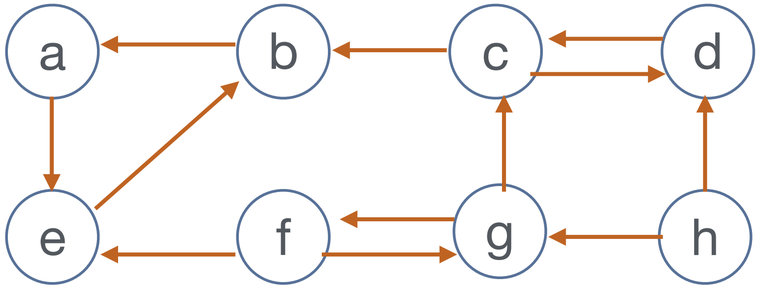
$b_{2,7}$

$a_{1,8}$

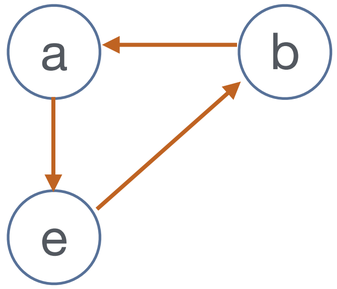
Nodes ordered by decreasing finishing times (popping-off order): a, b, e, c, g, f, d, h

Solution

G^T



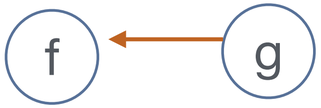
1st-round DFS
(start at a)



2nd-round DFS
(start at c)



3rd-round DFS
(start at g)



4th-round DFS
(start at h)



Acyclic component graph

